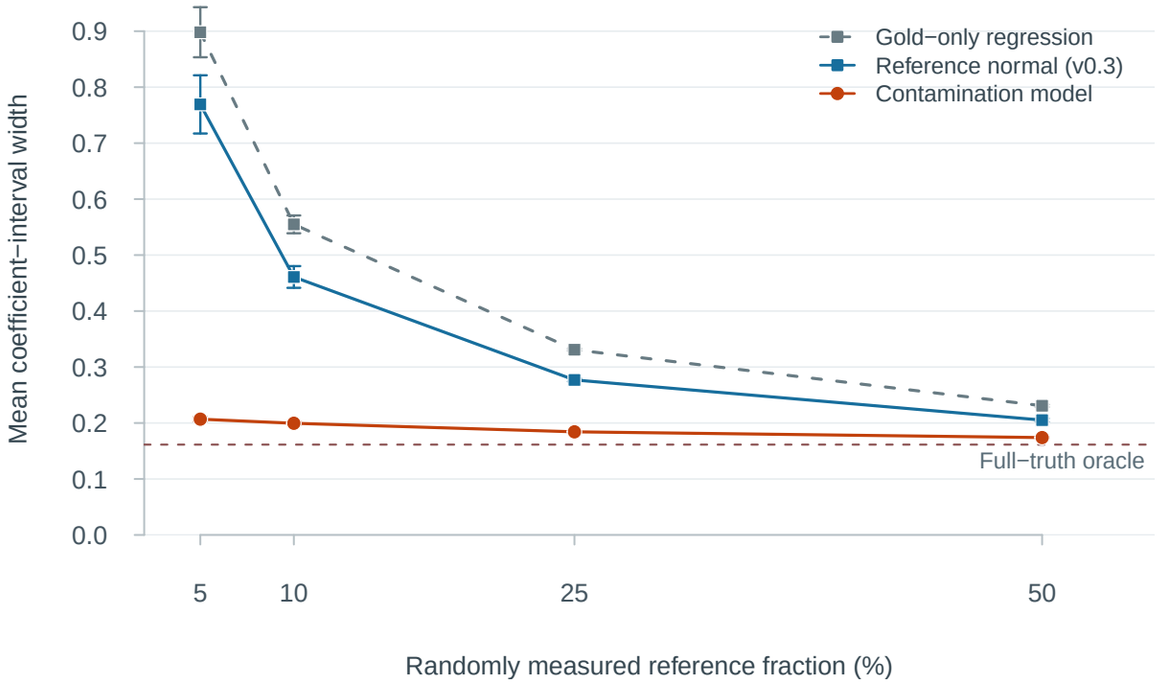
Using recorded values without trusting them

Held−out synthetic evidence | 20 imputations | A reference subset calibrates a model of recording errors

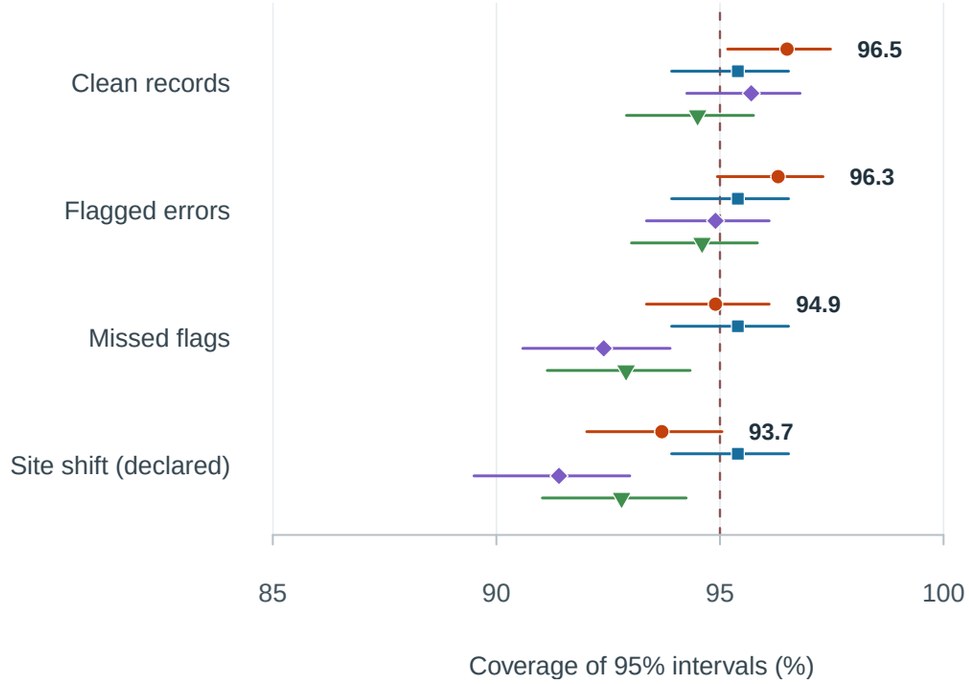
A Using recorded values buys precision

Missed flags. 25%: 1,000 cohorts; other fractions: 150 cohorts



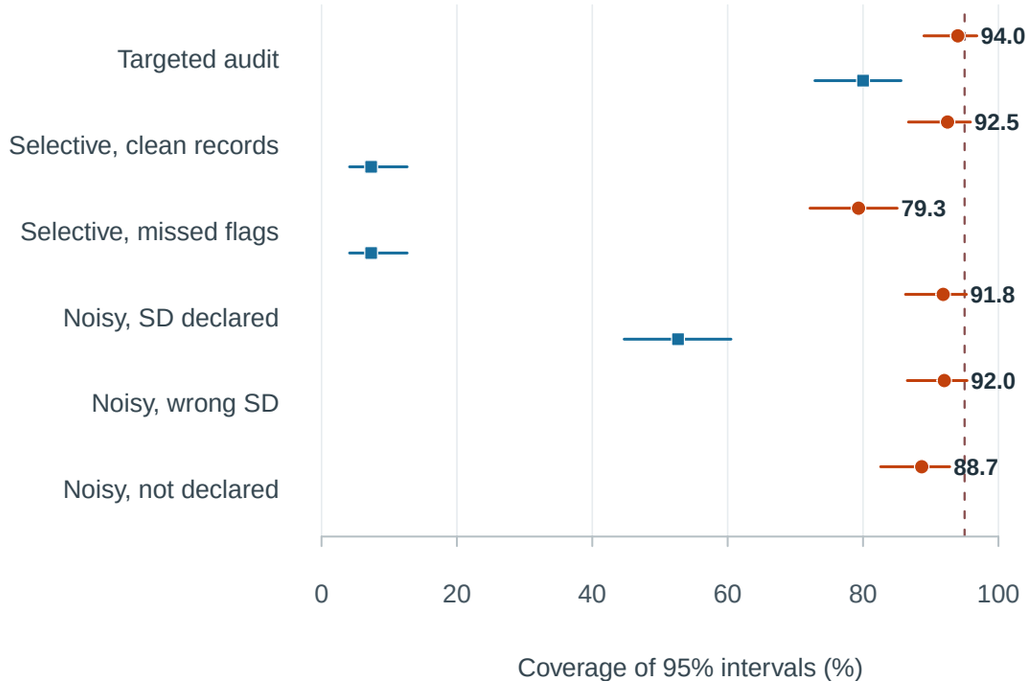
B Coverage where the model holds

Same data for every method; 1,000 cohorts per row. Errors match the model here, which favors it



C Selective or noisy references

25% expected references; 150 cohorts per row. Blank rows ran the contamination model alone



D Assumptions deliberately broken

Missed−flag fixture unless named; 150 cohorts per row. Coverage below 40% leaves the axis

Contamination model Reference normal (v0.3) Linear MI with recorded value Generalized raking

Coverage bars: 95% Wilson Monte Carlo intervals; dashed line: nominal 95%. Width bars: mean +/− 1.96 MCSE. Coverage and width are conditional on returned intervals.

Refusals and failures are in the study tables. Cells sharing a fixture reuse cohorts and are not independent replications. All data are synthetic.